Practice Exam, Weeks 2–7

Name (optional): _____

This is not for credit. It is a way to find out what you know before somebody else asks. Sixty-four questions, about forty minutes, closed notes and no model. Then mark it yourself from the key.

The graded concept checks are spoken and ask you to *define and apply* an idea in your own words. This paper only asks you to *recognize* it, which is easier. Treat a wrong answer here as a term to go and say out loud.

Part A · True or false 20 questions · circle T or F

1. A token is the same thing as a word. **T F**

2. "The dog bit the man" and "the man bit the dog" produce the identical bag of words. **T F**

3. Lemmatization always produces a real word; stemming need not. **T F**

4. A term that appears in every document of a corpus has an inverse document frequency of zero. **T F**

5. Cosine similarity gets larger when either of the two vectors gets longer. **T F**

6. word2vec gives the word "bank" a different vector in each sentence it appears in. **T F**

7. In skip-gram, the input is the surrounding words and the output is the one word in the middle. **T F**

8. Raising the temperature gives unlikely tokens a better chance of being chosen. **T F**

9. The outputs of softmax always add up to 1. **T F**

10. After training, a model keeps a copy of its corpus and looks answers up in it. **T F**

11. In one attention head, each word's row of attention weights adds up to 1. **T F**

12. The causal mask makes the weights on later words very small, but not zero. **T F**

13. Without positional encoding, attention could not tell "dog bites man" from "man bites dog". **T F**

14. Making a sequence twice as long roughly doubles the work attention has to do. **T F**

15. A bright cell in an attention heatmap shows which word caused the model's output. **T F**

16. A few-shot prompt changes the model's weights. **T F**

17. Wei et al. found that chain-of-thought prompting helped models of every size. **T F**

18. If every step in a chain of thought is correct, the final answer must be correct. **T F**

19. Under TruthfulQA's definition, "I have no comment" counts as a truthful answer. **T F**

20. Two markers who give the same mark to 90% of the answers must have a high Cohen's kappa. **T F**

Part B · Multiple choice 20 questions · one best answer each

1. A 200-word document contains the word "tiger" 10 times. Its term frequency for "tiger" is
   a. 10
   b. 0.5
   c. 0.05
   d. 20

2. TF-IDF is highest for a term that is
   a. frequent in this document and rare across the corpus
   b. frequent in every document
   c. rare in this document and rare across the corpus
   d. on the stop-word list

3. Which of these is a *dense* vector?
   a. a one-hot vector
   b. a row of bag-of-words counts
   c. a row of TF-IDF scores
   d. a word2vec embedding

4. The distributional hypothesis says that
   a. common words carry the most meaning
   b. words that appear in similar contexts have similar meanings
   c. every word has exactly one meaning
   d. word order does not affect meaning

5. Negative sampling made word2vec practical because it
   a. removes stop words before training
   b. gives each word several vectors
   c. avoids computing a softmax over the whole vocabulary at every step
   d. shrinks the vocabulary to a few thousand words

6. word2vec is trained to predict missing words. What is kept when training ends?
   a. its predictions for each sentence
   b. the corpus
   c. the sliding window
   d. the hidden-layer weights, which are the word vectors

7. The fixed-length bottleneck is the problem that
   a. a seq2seq encoder must squeeze the whole input into one context vector
   b. a vocabulary cannot grow after training
   c. attention scores every pair of words
   d. a prompt has a maximum length

8. BERT is best described as
   a. a decoder-only model trained to predict the next token
   b. an encoder-only model pretrained by recovering hidden words
   c. a counting method like TF-IDF
   d. a recurrent network with a hidden state

9. Which of these is an *architecture*?
   a. BERT
   b. few-shot prompting
   c. the transformer
   d. TF-IDF

10. In one attention head, the query and the key decide how much one word attends to another. The value is
   a. the position of the word in the sentence
   b. what the attended word hands over to be blended in
   c. the attention weight after softmax
   d. the word's one-hot vector

11. The steps inside one attention head, in order, are
   a. softmax, score, blend
   b. blend, score, softmax
   c. score, blend, softmax
   d. score, softmax, blend

12. In cross-attention,
   a. queries come from the decoder; keys and values come from the encoder
   b. queries, keys and values all come from the same sequence
   c. queries come from the encoder; keys and values come from the decoder
   d. two attention heads exchange their weights

13. The main reason the transformer displaced the recurrent network, apart from accuracy, is that it
   a. needs no training data
   b. has fewer parameters
   c. handles every position at once, which suits a GPU
   d. does not need tokenization

14. A chat model types its answer one token at a time because
   a. the interface is slowed down for readability
   b. each new token depends on the ones already written
   c. attention can only look at one word at a time
   d. temperature is set to zero

15. An ablation study
   a. tests a model on questions from outside its training data
   b. scores several models on the same benchmark
   c. adds worked examples to a prompt
   d. removes or replaces one part at a time and compares each version with the full one

16. The edit distance between “cat” and “cart” is
   a. 0
   b. 1
   c. 2
   d. 4

17. Which is the clearest example of an *imitative falsehood*?
   a. getting a long multiplication wrong
   b. declining to answer
   c. saying that cracking your knuckles causes arthritis
   d. misspelling an author's name

18. Adding “if unsure, say ‘I have no comment’” raises a prompt's truthfulness score from 7 to 14 out of 20. Before saying the prompt is better, the first thing to check is
   a. whether the model is larger
   b. how many of the answers are now refusals
   c. whether the questions were in alphabetical order
   d. the temperature

19. Two markers each pass 10 of 20 answers, and they give the same mark on 15. Their percent agreement is

a. 50%
b. 15%
c. 100%
d. 75%

20. On your 20-item evaluation set, prompt A scores 9 and prompt B scores 10. The best conclusion is that

a. one item is five points, which is not evidence of a difference; look at which items changed
b. prompt B is 5% better and should replace A
c. the evaluation set is broken
d. prompt A is more truthful

Part C · Matching 24 questions · write the letter; each letter is used once

Set 1 · How text becomes numbers

41. ___	**Corpus**	A. Very common words removed before counting
42. ___	**One-hot encoding**	B. The raw scores a model produces, one per token, before softmax
43. ___	**Stop words**	C. One word carrying several distinct meanings
44. ___	**N-gram**	D. The body of text a method learns from
45. ___	**Cosine similarity**	E. A run of adjacent words counted as one unit
46. ___	**Polysemy**	F. The fixed-size running summary a recurrent network carries forward
47. ___	**Hidden state**	G. A vector as long as the vocabulary containing a single 1
48. ___	**Logits**	H. A measure of the angle between two vectors, from −1 to 1

Set 2 · Attention and the transformer

49. ___	**Query**	A. Marks where in the sequence a token sits
50. ___	**Key**	B. Several sets of learned questions running side by side in one layer
51. ___	**Value**	C. The question a word asks of the other words
52. ___	**Multi-head attention**	D. Keeps a block's input by adding it back to what was just computed
53. ___	**Positional encoding**	E. The label a word wears, matched against other words' questions
54. ___	**Causal mask**	F. Lets the word being written look at the whole input that was read
55. ___	**Cross-attention**	G. What a word hands over to any word that attends to it
56. ___	**Add and normalize**	H. Stops a decoder from looking at words it has not written yet

Set 3 · Prompting, truthfulness and evaluation

57. ___	**Exemplar**	A. Agreement between two markers, corrected for chance
58. ___	**Benchmark**	B. Whether the steps a model writes are how it reached its answer
59. ___	**Out-of-distribution**	C. A worked example placed in a prompt before the question
60. ___	**Faithfulness**	D. The answer an evaluation treats as correct
61. ___	**Hallucination**	E. Actually addresses the question that was asked
62. ___	**Ground truth**	F. A fixed set of questions with known answers that everyone scores on
63. ___	**Informative**	G. A fluent, confident statement that is false
64. ___	**Cohen's kappa**	H. Unlike the pattern of the data a model was trained on

Answer Key

Mark it yourself. The entry in italics is the one to re-read.

Part A · True or false

1. **F.** Tokens are subword pieces; rare words split into several. *tokens and tokenization*
2. **T.** A bag of words discards order. *bag of words*
3. **T.** A lemma comes from a dictionary; a stem is whatever the rule leaves. *stemming, lemmatization*
4. **T.** log(N/N) = log 1 = 0. *inverse document frequency*
5. **F.** That is the dot product. Cosine divides the lengths out. *dot product, cosine similarity*
6. **F.** word2vec is static: one vector per word. *static and contextual embeddings*
7. **F.** That is CBOW. Skip-gram is the mirror image. *CBOW, skip-gram*
8. **T.** Higher temperature flattens the distribution. *temperature*
9. **T.** That is what makes them probabilities. *softmax*
10. **F.** Only the corpus's effects on the weights remain. *corpus, training and inference*
11. **T.** Softmax runs on each row. Attention is a budget. *attention scores and weights*
12. **F.** Minus infinity before softmax gives exactly zero. *masked attention*
13. **T.** Attention only scores pairs of words. *positional encoding*
14. **F.** Every pair is scored, so about four times. *what the transformer traded*
15. **F.** It shows how one head blended values. *attention weights are not explanations*
16. **F.** Only the text in front of the question changes. *zero-shot and few-shot prompting*
17. **F.** It made models under about ten billion parameters worse. *emergent abilities*
18. **F.** The mirage paper found right steps with wrong answers. *faithfulness of a chain of thought*
19. **T.** Truthful means not asserting anything false. *truthful versus informative*
20. **F.** If both say “wrong” to nearly everything, chance explains it. *inter-annotator agreement*

Part B · Multiple choice

1. **c.** 10 ÷ 200 = 0.05. *term frequency*
2. **a.** Frequent here, rare elsewhere. *TF-IDF*
3. **d.** The other three are long and mostly zeros. *sparse and dense vectors*
4. **b.** “The company it keeps.” *the distributional hypothesis*
5. **c.** A few random negatives instead of the whole vocabulary. *negative sampling*
6. **d.** The prediction task was only a means. *the pretext task, word2vec*
7. **a.** One vector, whatever the input's length. *the fixed-length bottleneck*
8. **b.** Encoder-only, masked language modeling. *BERT*
9. **c.** BERT is a model; the other two are techniques. *model, architecture and technique*
10. **b.** Query and key decide whether; the value is what. *query, key and value*
11. **d.** Dot products, then shares, then a blend of values. *attention scores and weights*
12. **a.** The decoder asks; the encoder's output answers. *cross-attention*
13. **c.** Parallel training is the column of Table 1 that mattered. *what the transformer traded*
14. **b.** Training is parallel; generation is not. *what the transformer traded, masked attention*
15. **d.** One part at a time, against a control. *ablation study*
16. **b.** Insert one letter. *exact match, edit distance and BLEU*
17. **c.** A popular myth is the likely continuation. *imitative falsehood*
18. **b.** A refusal is truthful and not informative. *truthful versus informative*
19. **d.** 15 ÷ 20. Chance would give 50%, so kappa is 0.5. *inter-annotator agreement*
20. **a.** One item is five points. *what twenty items can and cannot show*

Part C · Matching

Set 1: 41 D · 42 G · 43 A · 44 E · 45 H · 46 C · 47 F · 48 B
Set 2: 49 C · 50 E · 51 G · 52 B · 53 A · 54 H · 55 F · 56 D
Set 3: 57 C · 58 F · 59 H · 60 B · 61 G · 62 D · 63 E · 64 A

Reading your result

There is no pass mark. Count your misses by week and start with the week that has the most.

Material	Questions	Missed
Week 2 · text into numbers	A 1–10 · B 1–8 · C 41–48	/ 26
Week 3 · the transformer	A 11–15 · B 9–14 · C 49–56	/ 19
Week 6 · chain of thought	A 16–18 · B 15–16 · C 57–60	/ 9
Week 7 · truthfulness, evaluation	A 19–20 · B 17–20 · C 61–64	/ 10

For every miss, find the entry in the study guide and say its definition out loud without looking. Then try one of its “you should be able to” lines. That second step is what a concept check asks for.